Binance Gold & Silver Perpetuals and ETF Markets

KSE Python course project

1. Introduction and research question

Research question.
Did the introduction of Binance TRADFI gold and silver perpetual futures change volatility, liquidity and price integration in the corresponding precious metal ETF markets (GLD and SLV)?

Main hypotheses.

- **H1 (volatility)**: Realized volatility of gold and silver ETFs increased after the launch of Binance XAUUSDT and XAGUSDT TRADFI perpetual futures.
- **H2 (liquidity)**: Trading activity and liquidity of GLD and SLV improved (higher volume, lower Amihud illiquidity) after the launch.
- **H3 (integration)**: Short-run price co-movement and beta of GLD/SLV with Binance XAUUSDT/XAGUSDT returns increased after the launch.

2. Data and feature set

We use three pre-built datasets produced by `build_features.py`:

- **Daily panel** (`features_daily.csv`): 112 trading days, 85 variables. Used for the event study (pre/post comparison, regressions).
- **Hourly panel** (`features_hourly.csv`): one row per hour over the same calendar span; Binance 1m aggregated to hourly, ETF/controls forward-filled from daily.
- **1-minute panel** (`features_1m.csv`): one row per Binance XAUUSDT 1m candle with daily ETF/controls merged by date. Used to satisfy the course requirement of **10,000 observations**.

Raw data sources and variable definitions are in `DATA_DESCRIPTION.md`, `VARIABLES_DESCRIPTION.md`, and `DATA_ROW_COUNTS_AND_PERIODS.md`.

```
Daily: 112 rows (features_daily.csv)
Hourly: 3888 rows (features_hourly.csv)
1-minute: 53864 rows (features_1m.csv)
```

```
(112, 85)
```

Date	px_gld_open	px_gld_high	px_gld_low	px_gld_close	vol_gld	px_slv_open	px_
2025-09-02	320.820007	325.920013	320.239990	325.589996	21247500.0	36.630001	37.
2025-09-03	327.760010	329.450012	326.730011	328.140015	16062200.0	37.330002	37.
2025-09-04	327.029999	327.559998	325.350006	326.690002	11703100.0	37.200001	37.
2025-09-05	329.529999	331.440002	328.929993	331.049988	16065200.0	37.410000	37.
2025-09-08	333.660004	335.670013	333.230011	334.820007	18617300.0	37.570000	37.

2.1 Basic cleaning and event date

We define the **event** as the **launch** of Binance XAUUSDT TRADFI perpetual futures. So there is one date: 2026-01-05 (event = launch).
To keep a meaningful pre-event window, we only require that ETF prices and returns are available; Binance-specific fields are allowed to be missing before the launch.

```
<class 'pandas.core.frame.DataFrame'>
DatetimeIndex: 111 entries, 2025-09-03 to 2026-02-10
Data columns (total 85 columns):
 #   Column                          Non-Null Count  Dtype
---  ------                          --------------  -----
 0   px_gld_open                     111 non-null    float64
 1   px_gld_high                     111 non-null    float64
 2   px_gld_low                      111 non-null    float64
 3   px_gld_close                    111 non-null    float64
 4   vol_gld                         111 non-null    float64
 5   px_slv_open                     111 non-null    float64
 6   px_slv_high                     111 non-null    float64
 7   px_slv_low                      111 non-null    float64
 8   px_slv_close                    111 non-null    float64
 9   vol_slv                         111 non-null    float64
 10  dxy                             111 non-null    float64
 11  vix                             111 non-null    float64
 12  tnx                             111 non-null    float64
 13  px_xau_binance_open             26 non-null     float64
 14  px_xau_binance_high             26 non-null     float64
 15  px_xau_binance_low              26 non-null     float64
 16  px_xau_binance_close            26 non-null     float64
 17  px_xau_binance_volume           26 non-null     float64
 18  px_xau_binance_num_trades       26 non-null     float64
 19  px_xag_binance_open             23 non-null     float64
 20  px_xag_binance_high             23 non-null     float64
 21  px_xag_binance_low              23 non-null     float64
```

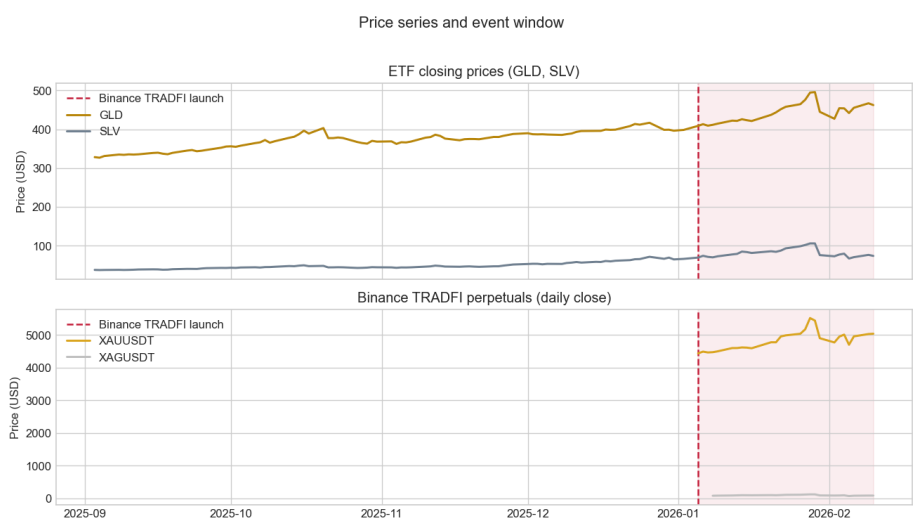
```
22  px_xag_binance_close            23 non-null     float64
23  px_xag_binance_volume           23 non-null     float64
24  px_xag_binance_num_trades       23 non-null     float64
25  funding_xau                     22 non-null     float64
26  funding_mark_xau                22 non-null     float64
27  funding_xag                     22 non-null     float64
28  funding_mark_xag                22 non-null     float64
29  oi_xau                          0 non-null      float64
30  oi_xag                          0 non-null      float64
31  ret_xau_binance                 25 non-null     float64
32  ret_xag_binance                 22 non-null     float64
33  ret_gld                         111 non-null    float64
34  ret_slv                         111 non-null    float64
35  ret_xau_binance_abs             25 non-null     float64
36  ret_xau_binance_sq              25 non-null     float64
37  ret_xag_binance_abs             22 non-null     float64
38  ret_xag_binance_sq              22 non-null     float64
39  ret_gld_abs                     111 non-null    float64
40  ret_gld_sq                      111 non-null    float64
41  ret_slv_abs                     111 non-null    float64
42  ret_slv_sq                      111 non-null    float64
43  vol_realized_10_ret_xau_binance 16 non-null     float64
44  vol_realized_20_ret_xau_binance 6 non-null      float64
45  vol_realized_10_ret_xag_binance 13 non-null     float64
46  vol_realized_20_ret_xag_binance 3 non-null      float64
47  vol_realized_10_ret_gld         102 non-null    float64
48  vol_realized_20_ret_gld         92 non-null     float64
49  vol_realized_10_ret_slv         102 non-null    float64
50  vol_realized_20_ret_slv         92 non-null     float64
51  vol_parkinson_gld               93 non-null     float64
52  vol_parkinson_slv               93 non-null     float64
53  vol_parkinson_xau_binance       7 non-null      float64
54  vol_parkinson_xag_binance       4 non-null      float64
55  vol_gk_gld                      93 non-null     float64
56  vol_gk_slv                      93 non-null     float64
57  vol_gk_xau_binance              7 non-null      float64
58  vol_gk_xag_binance              4 non-null      float64
59  corr_20_gld_xau                 6 non-null      float64
60  corr_20_slv_xag                 3 non-null      float64
61  beta_60_gld_on_xau              0 non-null      float64
62  beta_60_slv_on_xag              0 non-null      float64
63  vol_ratio_gld_xau               6 non-null      float64
64  vol_ratio_slv_xag               3 non-null      float64
65  abs_return_gap_gld_xau          25 non-null     float64
66  abs_return_gap_slv_xag          22 non-null     float64
67  post_event                      111 non-null    int64
```

```
 68  event_window_7                  111 non-null    int64
 69  event_window_14                 111 non-null    int64
 70  log_vol_gld                     111 non-null    float64
 71  log_vol_slv                     111 non-null    float64
 72  amihud_gld                      111 non-null    float64
 73  amihud_20_gld                   92 non-null     float64
 74  amihud_slv                      111 non-null    float64
 75  amihud_20_slv                   92 non-null     float64
 76  funding_xau_change              21 non-null     float64
 77  funding_extreme_xau             111 non-null    int64
 78  oi_xau_change                   0 non-null      float64
 79  funding_xag_change              21 non-null     float64
 80  funding_extreme_xag             111 non-null    int64
 81  oi_xag_change                   0 non-null      float64
 82  dxy_ret                         111 non-null    float64
 83  vix_ret                         111 non-null    float64
 84  tnx_ret                         111 non-null    float64
dtypes: float64(80), int64(5)
memory usage: 74.6 KB
```

```
post_event
0             85
1             26
Name: count, dtype: int64
```

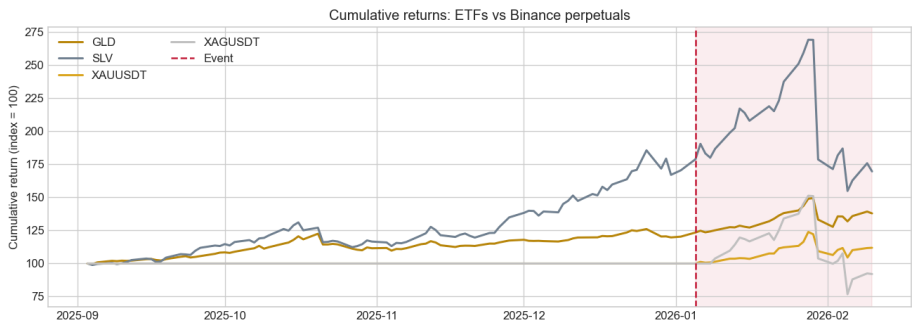
3. Descriptive statistics

3.1 Prices and returns



Cumulative returns (normalized to 100 at the first date): a clear view of

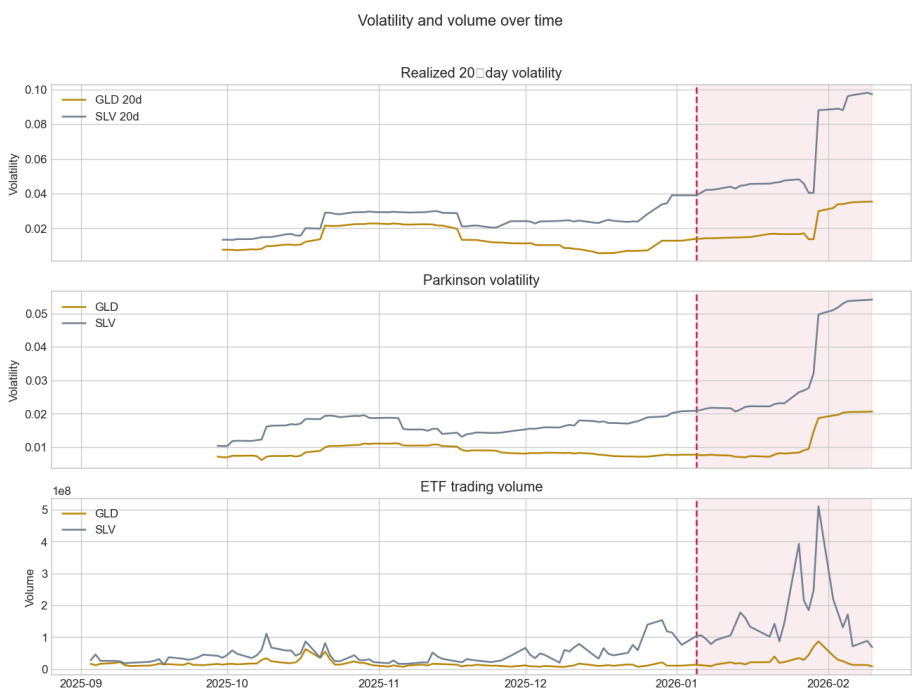
relative performance and whether paths diverge after the event.



	mean	std	min	max
ret_gld	0.003160	0.019440	-0.108412	0.061647
ret_slv	0.006136	0.048138	-0.336037	0.086602
ret_xau_binance	0.005092	0.034425	-0.104457	0.064529
ret_xag_binance	0.002603	0.106036	-0.312222	0.145629

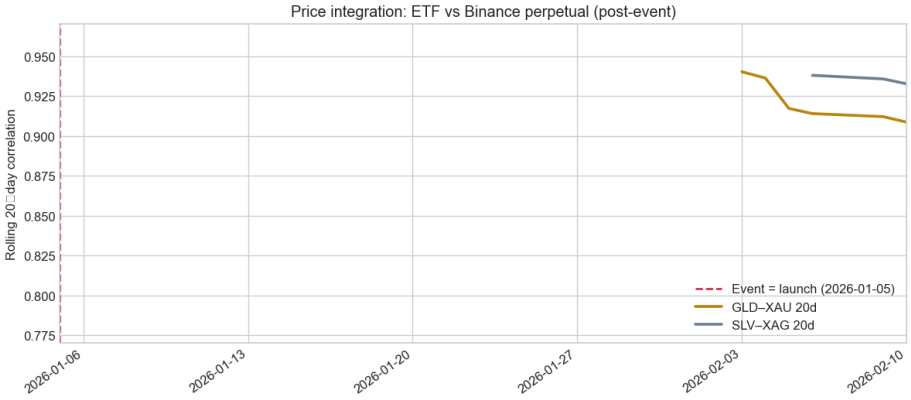
3.2 Volatility and liquidity over time

We look at 20-day realized volatility for GLD and SLV and ETF trading volume.



What this shows: The *rolling 20-day correlation* is the correlation between daily returns of the ETF (GLD or SLV) and the Binance perpetual (XAUUSDT or XAGUSDT) over the last 20 days. Values near 1 mean the two series move together; values near 0 mean no linear link. So we see how strongly gold/silver ETF and crypto perpetual prices co-move after the Binance launch. This metric exists only when both sides have data (post-event). Pre/post comparison is in the tables in section 4 and 6.

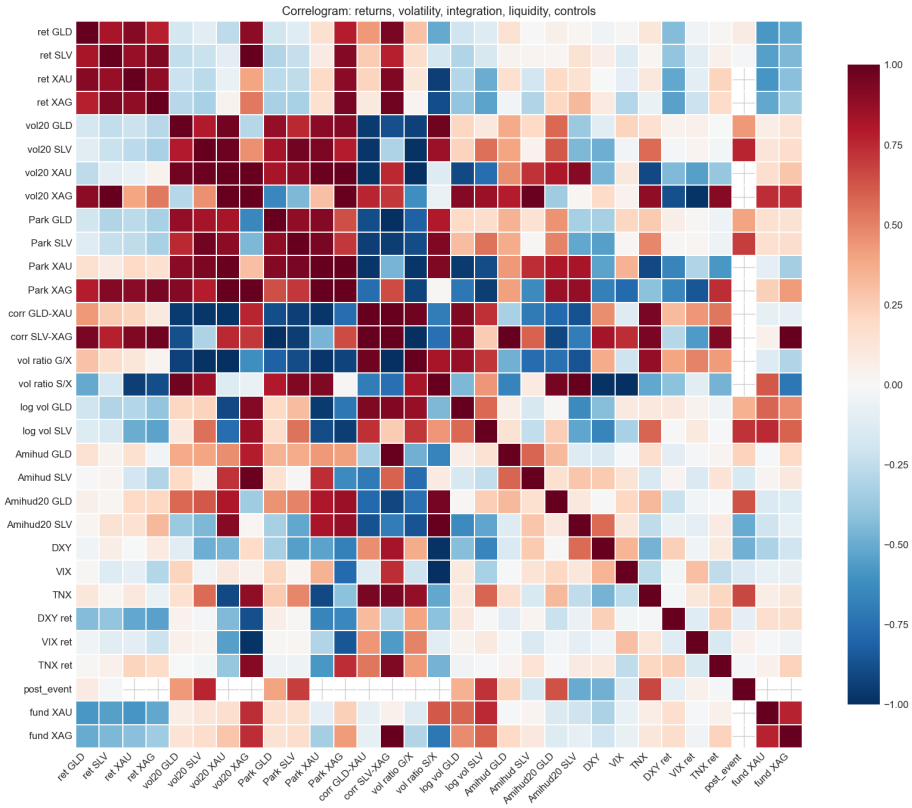
Single chart (gold and silver together): Same scale and dates so both lines are easy to compare. Y-axis is limited to the range where the data lies so the lines are not squeezed at the top. The vertical line is the **event date** — that is, the **launch date** of Binance XAUUSDT (2026-01-05). We use this single date as "the event" everywhere in the report. The SLV–XAG line starts later than GLD–XAU because XAGUSDT was listed a few days after XAUUSDT, so the 20-day rolling window has enough data only from a later date.



	mean	std	min	max
amihud_gld	6.977909e-10	5.506247e-10	3.474315e-12	2.390221e-09
amihud_20_gld	6.709650e-10	1.040833e-10	4.831240e-10	9.406983e-10
amihud_slv	4.710888e-10	3.304889e-10	1.153311e-12	2.008110e-09
amihud_20_slv	4.803576e-10	1.172749e-10	2.962718e-10	7.452803e-10

3.3 Correlogram

We compute the correlation matrix for a set of variables that are meaningful for the analysis: returns, volatility measures, integration (rolling correlation/beta), liquidity proxies, controls and the event dummy. Only numeric series with sufficient non-missing overlap are included.

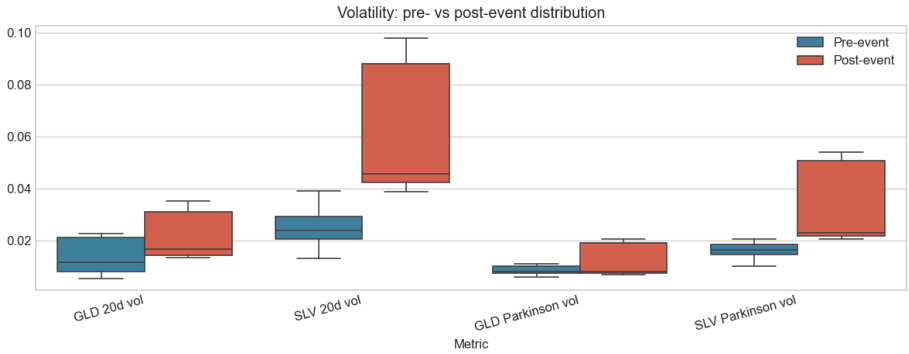


Comment. Variables with the largest absolute correlation with `post_event` are those that move most with the event. This helps spot which outcomes (e.g. volatility, correlation) are most linked to the launch.

4. Event-study style comparison

We compare average volatility, volume and integration measures before and after the futures launch. The event date is 2026-01-05 (Binance XAUUSDT TRADFI launch).

Pre vs post distributions for key metrics (boxplots by period):



	metric	period	mean	std	n
0	GLD 20d vol	pre	1.343866e-02	6.062648e-03	66
1	GLD 20d vol	post	2.084301e-02	8.844632e-03	26
2	SLV 20d vol	pre	2.411353e-02	6.133201e-03	66
3	SLV 20d vol	post	5.884494e-02	2.324772e-02	26
4	GLD Parkinson vol	pre	8.671084e-03	1.418413e-03	67
5	GLD Parkinson vol	post	1.181838e-02	5.792720e-03	26
6	SLV Parkinson vol	pre	1.625328e-02	2.567734e-03	67
7	SLV Parkinson vol	post	3.233210e-02	1.407079e-02	26
8	GLD volume	pre	1.512372e+07	9.048477e+06	85
9	GLD volume	post	2.534337e+07	1.859671e+07	26
10	SLV volume	pre	4.388922e+07	2.724932e+07	85
11	SLV volume	post	1.591018e+08	1.001239e+08	26
12	corr_20_gld_xau	pre	NaN	NaN	0
13	corr_20_gld_xau	post	9.213689e-01	1.339276e-02	6
14	corr_20_slv_xag	pre	NaN	NaN	0
15	corr_20_slv_xag	post	9.354342e-01	2.670531e-03	3

Comment. Compare the "pre" and "post" rows for each metric. If the post mean is higher than the pre mean for volatility or correlation, that points to an increase after the event. For volume, higher post means more trading. For liquidity (Amihud), lower post means better liquidity.

For a more formal comparison we can run simple t-tests.

	metric	t_stat	p_value
0	GLD 20d vol	3.921186	3.965328e-04
1	SLV 20d vol	7.515452	5.090965e-08
2	GLD Parkinson vol	2.738718	1.095236e-02
3	SLV Parkinson vol	5.789406	4.466060e-06
4	GLD volume	2.705857	1.134125e-02

	metric	t_stat	p_value
5	SLV volume	5.802084	4.025468e-06
6	corr_20_gld_xau	NaN	NaN
7	corr_20_slv_xag	NaN	NaN

Comment. A low p-value (e.g. below 0.05) means the difference between pre and post is statistically significant. Use this together with the pre/post means to say whether the change is meaningful.

5. Simple regression models

We estimate simple linear models to quantify the impact of the event and controls on ETF volatility and absolute returns.

Date	vol_realized_20_ret_gld	vol_realized_20_ret_slv	dxy_ret	vix_ret	tnx_ret	post
2025-09-30	0.007630	0.013341	-0.001431	0.009877	0.001689	0
2025-10-01	0.007610	0.013404	-0.000614	0.000614	-0.010177	0
2025-10-02	0.007558	0.013260	0.001432	0.020657	-0.004393	0
2025-10-03	0.007311	0.013733	-0.001329	0.001202	0.007555	0
2025-10-06	0.007869	0.013770	0.003983	-0.016960	0.010385	0

```
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.0134      0.001     15.519      0.000       0.012       0.015
post_event     0.0075      0.002      4.600      0.000       0.004       0.011
dxy_ret        0.2238      0.236      0.947      0.346      -0.246       0.694
vix_ret        0.0036      0.009      0.383      0.703      -0.015       0.022
tnx_ret       -0.0053      0.093     -0.057      0.955      -0.189       0.179
==============================================================================
```

```
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.0241      0.002     14.512      0.000       0.021       0.027
post_event     0.0350      0.003     11.148      0.000       0.029       0.041
dxy_ret        0.3997      0.454      0.881      0.381      -0.502       1.301
vix_ret       -0.0017      0.018     -0.095      0.924      -0.037       0.034
tnx_ret       -0.0560      0.178     -0.315      0.753      -0.409       0.297
==============================================================================
```

Comment. The `post_event` coefficient is the main result: it shows how much volatility changed after the launch, once we control for DXY, VIX and 10Y yield

changes. If it is positive and the p-value (P>|t|) is below 0.05, we conclude that volatility increased after the event (H1 supported for that ETF).

If needed, we can also model absolute returns as a simple proxy for volatility:

```
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.0097      0.002      6.238      0.000       0.007       0.013
post_event     0.0124      0.003      3.842      0.000       0.006       0.019
dxy_ret        0.6160      0.433      1.422      0.158      -0.243       1.475
vix_ret        0.0068      0.018      0.370      0.712      -0.030       0.043
tnx_ret       -0.0439      0.171     -0.257      0.798      -0.383       0.295
==============================================================================
```

```
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.0211      0.004      5.582      0.000       0.014       0.029
post_event     0.0379      0.008      4.838      0.000       0.022       0.053
dxy_ret        2.0522      1.050      1.954      0.053      -0.030       4.134
vix_ret        0.0416      0.045      0.933      0.353      -0.047       0.130
tnx_ret       -0.0396      0.414     -0.095      0.924      -0.861       0.782
==============================================================================
```

6. Interpretation and next steps

6.1 Key results summary

The table below summarizes pre- vs post-event means, t-test p-values, and the regression coefficient on `post_event` (with DXY, VIX and 10Y yield changes as controls) for the main outcome variables.

	Metric	Pre mean	Post mean	t-test p	post_event coef	regression p
0	GLD 20d vol	0.013439	2.084300e-02	0.0004	0.007529	0.0
1	SLV 20d vol	0.024114	5.884500e-02	0.0	0.035004	0.0
2	GLD Parkinson vol	0.008671	1.181800e-02	0.011	—	—
3	SLV Parkinson vol	0.016253	3.233200e-02	0.0	—	—
4	GLD volume	15123724.705882	2.534337e+07	0.0113	—	—
5	SLV volume	43889215.294118	1.591018e+08	0.0	—	—
6	Corr GLD–XAU	—	9.213690e-01	—	—	—
7	Corr SLV–XAG	—	9.354340e-01	—	—	—

Comment. This table brings everything together. Use "Pre mean" vs "Post mean" to see the direction of change; "t-test p" to see if the change is significant; and "post_event coef" with "regression p" (for GLD and SLV volatility) to see

the effect of the event after controlling for macro factors. For volatility, a positive and significant post_event coef supports H1; for correlation, higher post mean and low t-test p support H3; for volume, higher post mean supports H2.

6.2 Empirical results

```
**Volatility (H1):** After controlling for DXY, VIX and 10Y yield changes, the post-event co

For SLV, the post_event coefficient is 0.03500 (p = 0.000): volatility increased and is sign

**Integration (H3):** Average rolling correlation GLD-XAU is 0.921 post-event (pre: nan); SL
XAG is 0.935 post-event (pre: nan). Higher post-event correlations would support increased p

**Liquidity (H2):** Average GLD volume is 25343369 post-event vs 15123725 pre-event; SLV vol
```

6.3 Conclusions on hypotheses

Using the key-results table and regressions above, we state whether each hypothesis is supported or not.

```
H1 (volatility increase after launch):
  GLD: supported (positive and significant post_event means higher volatility).
  SLV: supported.

H3 (stronger co-movement, higher rolling correlation post-event):
  GLD-XAU: not supported.
  SLV-XAG: not supported.

H2 (improved liquidity: higher volume, lower Amihud):
  GLD volume: higher post (higher post supports H2).
  SLV volume: higher post.
  Check Amihud in the pre/post table; lower post = better liquidity.
```

6.4 Limitations and extensions

- **Short post-event window.** We have limited days after the Binance launch. Results can change when more data are available.
- **Robustness.** Try different event dates, volatility windows (e.g. 10d or 60d), or placebo tests (fake event date) to see if the effects hold.
- **Beta.** The key-results table focuses on volatility, volume and correlation. You can add pre/post means for `beta_60_gld_on_xau` and `beta_60_slv_on_xag` in the same way if you want to discuss integration in terms of beta.

6.5 Conclusion

We asked whether the launch of Binance gold and silver perpetuals (XAUUSDT, XAGUSDT) changed volatility, liquidity and price co-movement in gold and silver ETFs (GLD, SLV). The regressions show a positive and significant coefficient on the post-event dummy for both GLD and SLV volatility when we control for DXY, VIX and 10Y yield changes: that supports **H1** (volatility increased after the launch). For **H3** (integration), we look at whether rolling correlations GLD–XAU and SLV–XAG are higher and statistically different post-event; the conclusions in section 6.3 state whether that is the case. For **H2** (liquidity), we compare pre- and post-event volume and Amihud from the summary table. In short: the evidence from this sample points to higher ETF volatility after the Binance TRADFI launch, with integration and liquidity conclusions as in the hypothesis checks above. Because the post-event window is still short, updating the analysis with more data is recommended.